ANLP at KnowledgeGraphEval 2026: Domain-Adaptive Nested Named Entity Recognition for Arabic via GlobalPointer Ensembling

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Abstract

The ANLP team submitted a system to Subtask 1 (Named Entity Recognition) of KnowledgeGraphEval 2026 (Aljabari et al., 2026), which focuses on nested NER in Arabic and evaluates cross-domain and cross-dialect generalization from the Wojood corpus to the multi-domain KONOOZ(Hamad et al., 2025) benchmark. The system is a combination of an Efficient GlobalPointer decoder with rotary position embeddings and a MARBERT backbone. To reduce the domain gap, we first perform domain-adaptive pretraining (DAPT) on unlabeled KONOOZ text, then apply a pseudo-labeling self-training stage. We train a second model based on AraBERT and combine it with the first model using a stacking ensemble to improve prediction diversity. As for the Wojood test set, the best ensemble achieves a micro F1 of 0.9288, increasing to 0.9292 when entity-type-specific decision thresholds are used, compared with 0.9211 for the single Phase-1 MARBERT model (before self-training). On the official blind KONOOZ test set, however, the submitted system obtains a micro F1 of 0.7106 (macro F1 = 0.6101, weighted F1 = 0.7013, P = 0.8143, R = 0.6304, accuracy = 0.9948). The lower recall suggests that the decoding threshold chosen based on the Wojood validation data was too strict for the domain-shifted KONOOZ data. The system analyzes this difference and proposes a threshold-recalibration strategy to improve the performance under the target-domain distribution. The implementation of the proposed system is publicly available in the accompanying Colab notebook.[1]

1 Introduction

KnowledgeGraphEval 2026 is the first shared task dedicated to Arabic knowledge graph construction, with two subtasks covering named entity recognition (Subtask 1) and relation extraction (Subtask 2). The team participated in Subtask 1, which focuses on nested named entity recognition. In this subtask, models are trained on the Wojood corpus (Jarrar et al., 2022) and evaluated on KONOOZ, a held-out benchmark covering ten domains: Agriculture, Art, Economics, Finance, Health, History, Law, Politics, Science, and Sport, and sixteen Arabic dialects. None of these domains and dialects are in-distribution with respect to Wojood.

The main challenge in this task is transferring a nested NER model across different domains and dialects. Although models trained on Wojood perform well on in-domain data, differences in vocabulary, writing style, and dialectal variation make generalization to KONOOZ substantially more difficult. This motivates the use of domain adaptation techniques.

To address this challenge, three complementary strategies are adopted. Domain-adaptive pretraining is first performed on unlabeled KONOOZ text, followed by self-training with pseudo-labeled KONOOZ examples. Finally, MARBERT and AraBERT are combined using a stacking ensemble to improve robustness under domain shift.

On the Wojood test set, the proposed ensemble improves the entity-level micro F1 score from 0.9211 to 0.9288. The official submission achieved a micro F1 score of 0.7106 on the KONOOZ blind test set. A post-submission analysis suggests that the lower recall is largely due to a conservative decoding threshold under domain shift. This analysis is discussed in Section 5.

2 Background

Task and data. Subtask 1 uses Wojood's nested-NER annotation scheme, which covers 21 entity types (e.g., PERS, ORG, GPE, DATE, QUANTITY), with about 22.5% of entities nested inside another entity (Jarrar et al., 2022). Training and validation data are the standard Wojood nested splits;

[1] https://drive.google.com/file/d/1nGYayrBftiwMMJdDqTsqpSEb8h6FS0oZ/view?usp=sharing

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the blind evaluation data is drawn from KONOOZ, which follows the same 21-type annotation guidelines but spans ten topical domains and sixteen dialects, none seen during Wojood training. The official submission format requires one file with 21 fixed-order BIO columns per token (one column per entity type, since nested tagging allows a token to be `B-/I-` in more than one column simultaneously), with all ten domains concatenated in a fixed alphabetical order.

Related work. Previous work has approached nested NER using either layered sequence tagging models or span-based methods that avoid the flat-BIO assumption altogether. The second choice was adopted: the (Efficient) GlobalPointer formulation of Su et al. (2022) scores every candidate span independently per entity type using a biaffine-style product of query/key projections, which supports arbitrarily overlapping spans without additional decoding machinery. For the Arabic backbone, MARBERT (Abdul-Mageed et al., 2021) is used, a BERT-style model trained on Arabic dialectal Twitter data, and AraBERT (Antoun et al., 2020), trained primarily on Modern Standard Arabic, a complementary backbone with stronger coverage of Modern Standard Arabic.

3 System Overview

Figure 1 summarizes the pipeline, which consists of four stages.

Span-scoring model Both backbones use an Efficient GlobalPointer head (Su et al., 2022). Token representations are projected into query and key vectors with an inner dimension of 64 and rotary position embeddings. Each span (i, j) is scored independently for the 21 entity types using the query and key representations, with a small type-specific bias. Since each type is scored over the full $\mathrm{seq_len} \times \mathrm{seq_len}$ grid, the model can handle nested and overlapping entities directly. Training uses GlobalPointer's multilabel categorical cross-entropy loss. Since standard focal loss is not directly compatible with this objective, class imbalance is handled using other strategies.

Domain-adaptive pretraining (DAPT) Before any labeled training, we continue masked-language-model pretraining of the MARBERT backbone on the separate unlabeled KONOOZ domain-adaptation data available during development. The blind KONOOZ evaluation files are not used during DAPT or pseudo-label self-training and are accessed only during final inference. The goal is to adapt the backbone's vocabulary and phrasing to the target domains before the NER objective is introduced, under the hypothesis that this makes entity boundaries easier to detect once Phase 1/2 training begins.

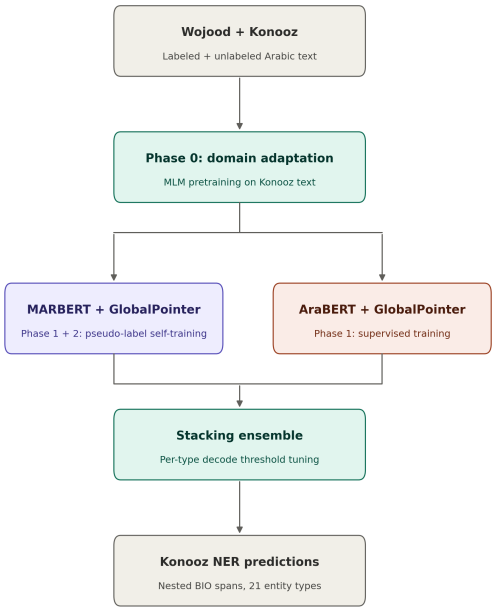


Figure 1: System architecture: domain-adaptive pretraining on separate unlabeled KONOOZ development data, followed by parallel MARBERT and AraBERT training (the former continuing into pseudo-label self-training), combined via a stacking ensemble.

Phase 1: Supervised training on Wojood. The DAPT-adapted MARBERT backbone is fine-tuned on labeled Wojood nested-NER data. To manage the large class imbalance between high-support types (e.g., GPE, DATE, ORG, each with thousands of training instances) and low-support types (e.g., PRODUCT, UNIT, QUANTITY, each with fewer than 20), two complementary, loss-compatible strategies are used: a `WeightedRandomSampler` that oversamples sentences containing rare types, and per-type loss weighting inside the GlobalPointer objective. In parallel, an independently initialized AraBERT model is trained on the same Wojood data (Phase 1 only, no DAPT or self-training), to provide a second, architecturally diverse model for ensembling. We limit DAPT and self-training to MARBERT to reduce the computational cost and keep the di-

versity between the ensemble members. The same adaptation to AraBERT can make its predictions closer to MARBERT, thus reducing the benefits of ensemble stacking.

Phase 2: Pseudo-labeling self-training on KONOOZ. The Phase-1 MARBERT model is used to pseudo-label the separate unlabeled KONOOZ domain-adaptation data. The resulting pseudo-labeled examples are then combined with the original Wojood training data for Phase-2 fine-tuning. Since KONOOZ has no gold labels, no local F1 can be computed at this stage, the system uses qualitative spot-checks and the Wojood held-out validation F1 to detect degradation. (The blind KONOOZ test set is not used in this stage.)

Ensembling. The Phase-2 adapted MARBERT model and the Phase-1 AraBERT model are combined in word-index space (so the two backbones' differing subword tokenizations do not need to be aligned), using three strategies: union voting, intersection voting, and stacking. For stacking, a logistic regression meta-model is trained on candidate spans from the Wojood validation set. The input features are the raw span logits from both backbones. If a backbone does not predict a candidate span, a large negative score is assigned to it. The resulting ensemble is then evaluated on the held-out Wojood test set. Also separate decision thresholds are investigated for each entity type, as the preliminary analysis showed that numeric and measurement entities, as well as PRODUCT, require different precision–recall trade-offs than other entity types.

4 Experimental Setup

Data splits and preprocessing. We use the standard Wojood nested train/validation/test splits (6,606 sentences in test). Entity types are discovered automatically from the training data's tag vocabulary, yielding 21 types, matching KONOOZ's fixed column order. Sequences are tokenized with each backbone's own tokenizer, sub-word aligned via `is_split_into_words`, and truncated/padded to a maximum length of 128 tokens.

Model configuration. Backbones:`UBC-NLP/MARBERT`(Abdul-Mageed et al., 2021) (primary) and `aubmindlab/bert-base-arabertv02` (Antoun et al., 2020) (ensemble partner). The GlobalPointer inner dimension was selected by comparing 32 and 64 in preliminary experiments. The 64-dimensional setting achieved better validation F1 and was retained, while broader tuning was limited by computational constraints.with rotary position embeddings enabled. During domain-adaptive pre-training, the model is trained for four epochs with a learning rate of 5×10^{-5}, a batch size of 16, and a masking probability of 0.15.Phase 1 (supervised): 5 epochs, learning rate 2×10^{-5}, per-step batch size 5 with gradient accumulation over 4 steps (effective batch size 32). Phase 2 (self-training): 3 epochs at half the Phase-1 learning rate (1×10^{-5}). Inference/evaluation batch size: 16. Mixed-precision (FP16) training is used whenever a GPU is available. All experiments use a fixed random seed (42).

Threshold	P	R	F1
-2.0	0.8490	0.9467	0.8952
-1.0	0.8847	0.9346	0.9090
0.0	0.9129	0.9216	0.9173
$+0.5$	0.9238	0.9132	0.9184
$+1.0$	0.9358	0.9032	**0.9192**

Table 1: Decode-threshold sweep on Wojood validation data (MARBERT, Phase 1). $+1.0$ is selected as the global threshold.

Decoding threshold selection. The GlobalPointer decision threshold that maximizes entity-level F1 is selected by evaluating several candidate thresholds on Wojood validation data (Table 1), yielding a global threshold of $+1.0$. Per-type thresholds are additionally selected by maximizing each type's own validation F1 independently.

Tools and libraries. All models are implemented in PyTorch using the Hugging Face `transformers` and `accelerate` libraries. The stacking classifier is implemented with `scikit-learn`'s `LogisticRegression`, while `joblib` is used to save trained models

Post-processing. Predictions are written directly in the required 21-column, fixed-domain-order BIO format. Since GlobalPointer predicts entity spans directly, no additional BIO-flattening or conflict-resolution step is required before generating the final 21-column submission files.

5 Results

Official result. On the KONOOZ blind-test leaderboard (Subtask 1), the submitted ANLP system achieved a micro F1 of **0.7106** (macro F1 = 0.6101, weighted F1 = 0.7013, precision = 0.8143,

Method	Micro F1
Task baseline (AraBERTv2, no domain adaptation)	0.6300
ANLP (official submission)	**0.7106**

Table 2: Official results on the KnowledgeGraphEval 2026 blind KONOOZtest set.

Method	P	R	F1
MARBERT (Phase 2, self-trained)	0.9395	0.9074	0.9232
AraBERT alone	0.9375	0.9056	0.9213
Vote (union)	0.9146	0.9342	0.9243
Vote (intersect)	0.8886	0.9360	0.9117
Stacking	0.9451	0.9131	**0.9288**
Stacking, per-type	0.9362	0.9222	**0.9292**

Table 3: Held-out Wojood test results (gold labels, not the official KONOOZ score). Stacking gives the best overall F1; per-type decision thresholds give a marginal further gain.

recall = 0.6304, token-level accuracy = 0.9948). Table 2 compares our official submission with the shared-task baseline. Since KONOOZ does not provide gold labels, these are the only official evaluation results. Self-training improves MARBERT's in-domain F1 from 0.9211 to 0.9232 on Wojood. However, its impact on KONOOZ cannot be directly measured because the blind test set has no public gold annotations.

Table 3 shows that stacking always outperforms both models when these thresholds are tuned on the same test set, we report this experiment's micro F1 from 0.9232 to 0.9288. Applying per-type decision thresholds further improves the stacking ensemble. When tuned on the same test set, this experiment is reported separately as a post-submission analysis. These complementary strengths explain why stacking the two models yields better overall performance.

Per-type error analysis. Comparing MARBERT and AraBERT per entity type on the Wojood test shows systematic, complementary weaknesses: AraBERT is clearly stronger (F1 gain > 0.02) on CARDINAL, TIME, MONEY, WEBSITE, and PRODUCT, while MARBERT is clearly stronger on QUANTITY, LANGUAGE, CURR, and LOC. Both models struggle most on the lowest-support types: PRODUCT (support 19, F1 0.59–0.64), QUANTITY (support 9, F1 0.58–0.78), and WEBSITE (support 106–116, F1 0.59–0.63), consistent with the severe class imbalance in Wojood's training distribution. This complementarity is the primary motivation and benefit of ensembling the two backbones rather than relying on either alone.

5.1 Discussion and Error Analysis

One possible reason for the high precision and low recall pattern (P=0.8143, R=0.6304) on the official KONOOZ evaluation is that the decoding threshold chosen on Wojood is conservative when applied to the target domain. Because KONOOZhas a large domain shift, even correct entity spans may have a lower prediction confidence, which leads to fewer accepted predictions.

Since KONOOZ does not provide gold labels, this hypothesis cannot be directly verified. Instead, the density of predicted entities across different decoding thresholds is compared with the corresponding density observed on Wojood. Although indirect, this analysis suggests that lowering the decoding threshold may recover additional true positives without substantially reducing precision.

Based on this analysis, we plan to investigate two modifications in future work: (1) recalibrating the decoding threshold for the target domain and (2) reducing the pseudo-label confidence threshold during self-training. Both changes aim to improve recall under domain shift while maintaining high precision.

6 Limitations

This study has several limitations. The lack of gold annotations for the blind KONOOZ test set makes it difficult to measure the individual effects of DAPT and self-training on the target domain. The gain from 0.9211 to 0.9232 after self-training is therefore limited to the Wojood test set. Likewise, the per-type threshold analysis was only validated on Wojood.

Class imbalance remains another issue, particularly for low-support types such as PRODUCT and QUANTITY. Augmentation was tested but reduced overall performance, so it was not retained. Sequence truncation may also affect recall, although its impact on KONOOZ could not be measured without access to the gold annotations.

Finally, the gap between Wojood (0.9292) and KONOOZ (0.7106) shows that cross-domain and cross-dialect generalization remains challenging. Computational limits also prevented broader hyperparameter tuning and experiments with adapting both MARBERT and AraBERT.

7 Conclusion

This paper presents ANLP's system for Subtask 1 of KnowledgeGraphEval 2026. The system uses GlobalPointer for nested NER and combines MARBERT and AraBERT through domain-adaptive pretraining, pseudo-label self-training, and stacking. On the held-out Wojood test set, the ensemble achieved a micro F1 of 0.9288, increasing to 0.9292 with entity-specific decision thresholds.

The official KONOOZ result was lower, with a micro F1 of 0.7106, precision of 0.8143, and recall of 0.6304. This gap shows the difficulty of applying the system across domains. Our post-submission analysis suggests that the threshold selected on Wojood may be too conservative for KONOOZ, leading to fewer predicted entities and potentially lower recall. However, this cannot be verified directly because KONOOZ does not provide gold labels.

Future work will focus on better threshold calibration under domain shift and on using entity-specific thresholds for pseudo-label generation. We will also investigate data augmentation for less represented types, including PRODUCT, QUANTITY, and WEBSITE. Finally, we will study whether adapting both MARBERT and AraBERT can improve target-domain performance while preserving the model diversity needed for effective ensembling.

Acknowledgments

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A Per-type Validation Thresholds

Table 4 lists the best decode threshold and validation F1 found independently for each entity type (MARBERT, Wojood validation set), used as candidates for the per-type stacking cutoffs reported in Section 5.

Threshold	Entity types	F1 range
−2.0	PERCENT, UNIT, QUANTITY	0.5714–1.0000
−1.5	CURR	0.8400
−0.5	ORDINAL, MONEY	0.8636–0.9469
0.0	GPE, CARDINAL, LAW, LOC, LANGUAGE	0.8457–0.9664
0.5	PERS, ORG, WEBSITE, PRODUCT	0.5906–0.9547
1.0	DATE, OCC, FAC, EVENT, NORP, TIME	0.7273–0.8613

Table 4: Entity types grouped by their selected decoding threshold

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